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PAPER_1104: Unified $F_{U,Bi}$ for SMBH Binary Merger Dynamics

Abstract

We develop the complete buoyancy-mediated unified field theory for supermassive black hole (SMBH) binary mergers, distinguishing inside-to-outside (inspiral → ringdown) and outside-to-inside (environment → binary) dynamics. The two directional $F_{U,Bi}$ components are derived from the 26-layer UQFF gravity framework with gravitational wave phonon coupling and chirp mass scaling.

1. Inside-to-Outside: Inspiral to Ringdown

The $F_{U,Bi}$ component radiating outward during the merger:

$$F_{U,Bi}^{\text{in}\to\text{out}} = \sum_{i=1}^{26} [\text{Ug1}_i + \text{Ug2}_i + \text{Ug3}_i + \text{Ug4}_i] \times \left(1 + \frac{\Phi}{F_U} \cdot \Delta E_{\text{gw}}\right) \times \left(\frac{\mathcal{M}_c}{M_{\text{total}}}\right)^{5/3}$$

26-Layer Gravity Terms

Each layer $i \in [1, 26]$ has quantum state factor $Q_i = 1/(1 + \kappa i)$:

Term	Expression	Physics
Ug1_i	$Q_i \cdot \mu_s \nabla(M_s/r)$	DPM-seeded gravity
Ug2_i	$(1 - [\text{SSq}])Q_i \cdot \mu_s \nabla(M_s/r) \times 0.1$	Charge-reactivity
Ug3_i	$[\text{SSq}]\, Q_i \cdot \mu_s \nabla(M_s/r) \times 0.05 \sin i$	String rotation
Ug4_i	$\kappa\, Q_i \times 10^{-15}$	Vacuum concentration

Chirp Mass Factor

For a binary with masses M_1 and M_2:

$$\mathcal{M}_c = M_{\text{total}} \cdot \eta^{3/5}, \qquad \eta = \frac{M_1 M_2}{M_{\text{total}}^2}$$

$$\left(\frac{\mathcal{M}_c}{M_{\text{total}}}\right)^{5/3} = \eta$$

GW Phonon Coupling

$$\Delta E_{\text{gw}} = \frac{G M_1 M_2}{2 r_{\text{sep}}}$$

The phonon-GW coupling factor:

$$1 + \frac{\Phi(\omega, \Gamma)}{F_U} \cdot \Delta E_{\rm gw}$$

2. Outside-to-Inside: Environmental Buoyancy

The complementary $F_{U,Bi}$ component from the environment:

$$F_{U,Bi}^{\rm out\to in} = F_U - F_{U,Bi}^{\rm in\to out} - \sum_k F_{{\rm tidal},k}(r_k, M_k)$$

where $F_{{\rm tidal},k}$ are tidal forces from surrounding structures (galaxy clusters, dark matter halos, accretion disk).

3. Net Buoyancy Imbalance

$$\Delta F = F_{U,Bi}^{\rm in\to out} - F_{U,Bi}^{\rm out\to in}$$

- $\Delta F > 0$: merger-dominated (radiative outflow exceeds environmental inflow)
- $\Delta F < 0$: environment-dominated (accretion exceeds radiation)
- $\Delta F = 0$: buoyancy equilibrium

4. Merger Timescale

Using the Peters formula for gravitational wave inspiral:

$$t_{\rm merger} = \frac{5}{256} \frac{c^5 a_0^4}{G^3 M_1 M_2 M_{\rm total}}$$

5. Implementation

Calculator: `UnifiedFUBiSMBHMergerDynamicsCalculator` in CondensedPhysics.py

- Inputs: M_1, M_2, $r_{\rm sep}$, F_U, tidal forces, linewidth Γ
- Outputs: both $F_{U,Bi}$ components, net imbalance, chirp mass, merger timescale

6. Conclusion

The directional decomposition of $F_{U,Bi}$ during SMBH mergers provides a complete description of buoyancy flow in the UQFF framework, with the inside-to-outside component governed by chirp mass scaling and GW phonon coupling, and the outside-to-inside component determined by the global unified field budget less tidal contributions.

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